1) **Assertion (A) :** When rays of white light pass through a prism, on emerging they give spectrum of seven colours.

Reason (R) : It is due to the scattering of light that red light bends minimum and violet light bends the maximum.

2) **Assertion (A) :** Myopic eye cannot see distant objects distinctly.

Reason (R) : For the correction of myopia converging lenses of appropriate power are prescribed by eye-surgeons.

3) **Assertion (A) :** When ciliary muscles contract, eye lens becomes thin.

Reason (R) : Ciliary muscles control the power of the eye lens.

4) **Assertion (A) :** The rainbow is a natural spectrum of sunlight in the sky.

Reason (R) : Rainbow is formed in the sky when the sun is overhead and water droplets are also present in air.

5) **Assertion (A) :** Red light signals are used to stop the vehicles on the road.

Reason (R) : Red coloured light is scattered the most so as to be visible from a large distance.

(A) Both, Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both, Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

6) A thin beam of white light falls on a face of a glass prism and a spectrum is obtained on a screen. Which of the following options is the correct sequence of colours from top to bottom on the screen ?

(A) Red, Green, Yellow, Violet, Blue
(B) Red, Yellow, Green, Blue, Violet
(C) Violet, Blue, Green, Yellow, Red
(D) Violet, Red, Green, Yellow, Blue

7) When you look at an object very close to your eyes, the :

(A) Ciliary muscles of your eye contract and the eye lens becomes thick.

(B) Ciliary muscles of your eye get relaxed and the eye lens becomes thick.

(C) Ciliary muscles of your eye contract and the eye lens becomes thin.

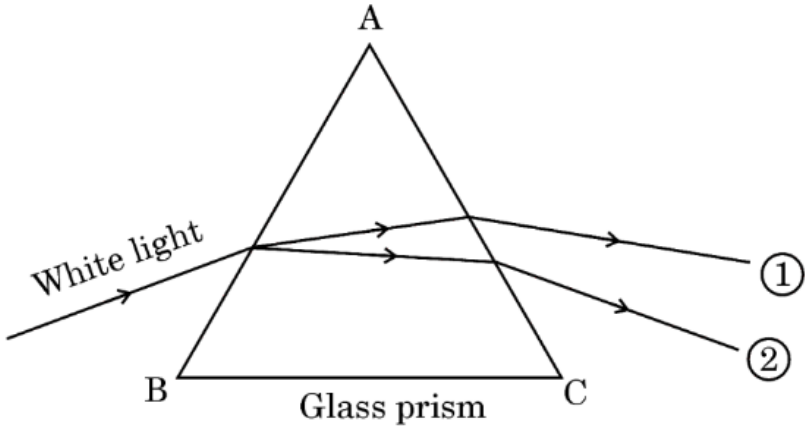
(D) Ciliary muscles of your eye get relaxed and the eye lens becomes thin.

8) The pair of eye parts responsible for admitting different amount of light in to the eyes is

(A) Iris and pupil (B) Ciliary muscles and pupil

(C) Retina and Iris (D) Ciliary muscles and cornea

9) Observe the given ray diagram showing refraction through a prism and choose the correct option for colour of rays 1 and 2 :



	Colour of ray 1	*Colour of ray 2*
(A)	Blue	Red
(B)	Green	Yellow
(C)	Red	Violet
(D)	Violet	Red

10) The vision defect which arises due to gradual weakening of the ciliary muscles and diminishing flexibility of eye lens is :

(A) Myopia (B) Hypermetropia

11) (C) Presbyopia (D) Myopia and hypermetropia both

To restore clear vision in persons whose size of the eye ball has reduced, he/she is suggested to use suitable

(A) Converging lens (B) Diverging lens

(C) Bifocal lens (D) Cylindrical lens

12) An old person is suffering from an eye defect caused by weakening of ciliary muscles and diminishing flexibility of the eye lens. If the defect of vision is 'a' which can be corrected by lens 'b', then 'a' and 'b' respectively are :

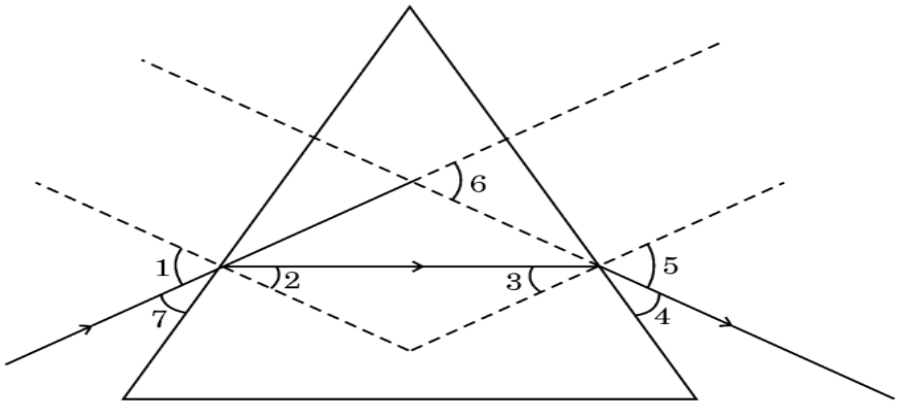
(A) hypermetropia and convex lens

(B) presbyopia and bifocal lens

(C) myopia and concave lens

(D) myopia and bifocal lens

13) In the given figure the angle of incidence and the angle of deviation respectively are :



14) When a beam of white light passes through a region of very fine dust particles, the colour of light that scatters the most in that region is :

(A) red (B) orange

(C) blue (D) yellow

15) The possible way to restore clear vision of those people whose eyeball has elongated is the use of suitable

(A) bifocal lens (B) concave lens

(C) converging lens (D) convex lens

16) The part of human eye which controls the amount of light entering into it.

(A) Iris (B) Cornea

(C) Ciliary muscles (D) Pupil

17) The lens system of human eye forms an image on a light sensitive screen, which is called as :

(a) Cornea

(b) Ciliary muscles

(c) Optic nerves

(d) Retina

18) Consider the following statements in the context of human eye :

(a) The diameter of the eye ball is about 2.3 cm.

(b) Iris is a dark muscular diaphragm that controls the size of the pupil.

(c) Most of the refraction for the light rays entering the eye occurs at the crystalline lens.

(d) While focusing on the objects at different distances the distance between the crystalline lens and the retina is adjusted by ciliary muscles.

The correct statements are –

(A) (a) and (b) (B) (a), (b) and (c)

(C) (b), (c) and (d) (D) (a), (c) and (d)

19) The Phenomena of light involved in the formation of a rainbow in the sky are

(A) Refraction, dispersion and reflection

(B) Refraction, dispersion and total internal reflection

(C) Dispersion, scattering and reflection

(D) Dispersion, refraction and internal reflection

20) When a beam of white light passes through a region having very fine dust particles, the colour of light mainly scattered in that region is :

(A) Red (B) Orange

(C) Blue (D) Yellow

21) What is the light-sensitive screen inside the human eye on which the image is formed?

22) Define the power of accommodation of the eye.

23) What is the least distance of distinct vision for a young adult with normal vision?

24) What is the far point of a normal human eye?

25) Name the dark muscular diaphragm behind the cornea that controls the size of the pupil.

26) What eye defect is also known as nearsightedness?

27) Which type of lens is used to correct myopia?

28) Define dispersion of light.

29) What is the acronym used to remember the sequence of colours in the spectrum of white light?

30) Name the phenomenon responsible for the twinkling of stars.